

Iteration in Programming

while loops

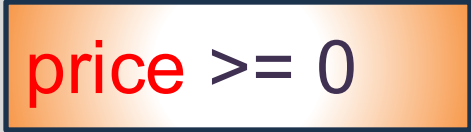
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Topics list

1. Repetition in Programming – Intro to looping
2. Use of loops (while loops).

Recap: Boolean conditions

- A boolean condition is an expression that evaluates to either true or false e.g.



price >= 0

- Boolean conditions can be used to control:
 - Selection i.e. if statements and
 - Iteration i.e. loops (we will look at these now).

Repetition in Programming

- Computers are very good at repetition.
- Example:
 - `calculate pay` for 1000 employees.
 - You should use the same `calculate pay` algorithm 1000 times.
 - You don't write the `calculate pay` algorithm 1000 times; instead you include it in a loop.

Repetition without loop

- Print “Hello world” to the screen 4 times

```
System.out.println("Hello World");  
System.out.println("Hello World");  
System.out.println("Hello World");  
System.out.println("Hello World");
```

```
NoLoop.main({ });  
Hello World  
Hello World  
Hello World  
Hello World
```

Introducing loop form

- Print “Hello world” to the screen 4 times
 - With a loop:
 - do this 4 times

```
System.out.println("Hello World");
```

```
NoLoop.main({ });|  
Hello World  
Hello World  
Hello World  
Hello World
```

Introducing loop form

- Print “Hello world” to the screen 4 times
- Introduce a counter

Set a counter to 1

do this while counter $\leq$ 4

```
System.out.println("Hello World");
```

increase counter by one

```
NoLoop.main({ });|  
Hello World  
Hello World  
Hello World  
Hello World
```

Topics list

1. Repetition in Programming – Intro to looping
2. Use of loops (while loops).

Loops in Programming

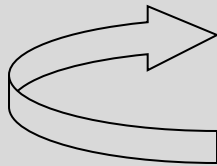
- There are three types of loop in (Java) programming:
 - **while** loops
 - **for** loops
 - **do while** loops

while loop pseudo code

General form of a while loop

while keyword

boolean condition



```
while(loop condition) {  
    loop body  
}
```

Statements to be repeated



Pseudo-code expression of the actions of
a while loop

while we wish to continue, do the things in the loop body

Construction of while loop

Declare and initialise **loop control variable (LCV)**

while(boolean condition based on **LCV** is true)

{

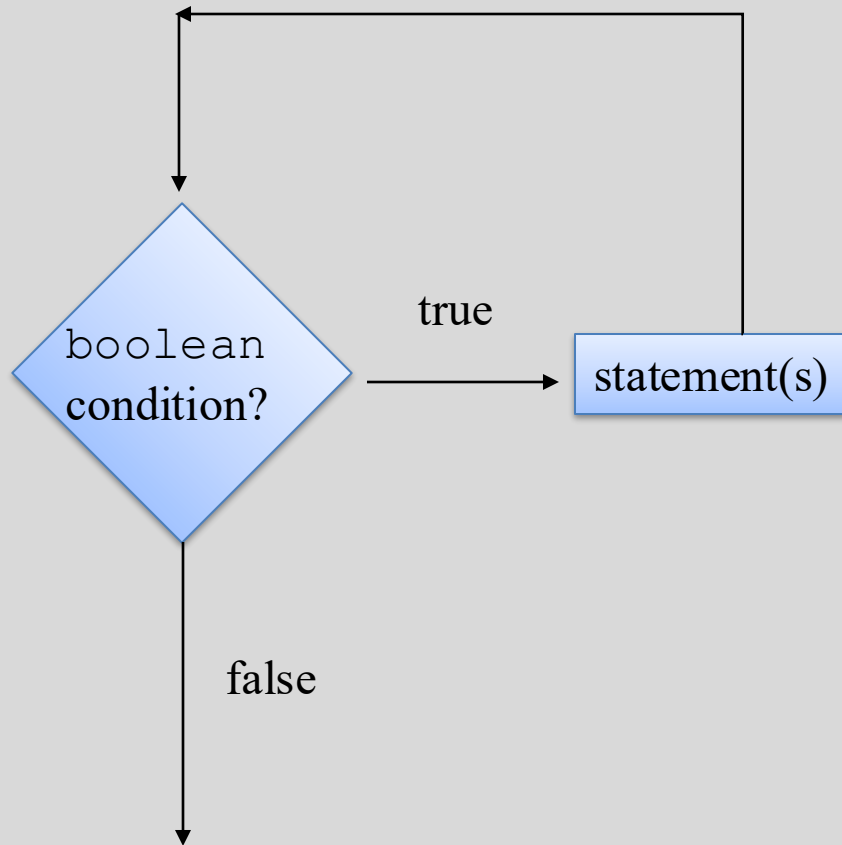
 “do the job to be repeated”

 “update the **LCV**”

}

This structure should always be used

while loop Flowchart



```
int i = 1;
while (i <= 4)
{
    System.out.println("Hello World");
    i++;
}
```

```
NoLoop.main({ });|
Hello World
Hello World
Hello World
Hello World
```

Some Study Exercises

1. Change the code so that “Hello World” is printed out 10 times.
2. Change the code so that the numbers from 1 to 10 (inclusive) are printed out, one line at a time.
3. Change the code so that the numbers from 10 to 1 are printed out.

Loop using the counter variable

```
int i = 1;

while (i <= 6) {
    System.out.println("Num : " + i);
    i++;
}
```

```
ForNumberLoop.main({ });
```

```
Num : 1
```

```
Num : 2
```

```
Num : 3
```

```
Num : 4
```

```
Num : 5
```

```
Num : 6
```

Questions?

